A Deep Multiscale Pyramid Network Enhanced With Spatial–Spectral Residual Attention for Hyperspectral Image Change Detection

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***Abstract*— Change detection plays an important role in Earth surface observation and has been extensively investigated over recent decades. A hyperspectral image (HSI) with high spectral resolution provides abundant ground object information, which is expected by finer change detection. The existing convolutional neural network (CNN)-based methods extract image features with a fixed kernel, which is incompetent to cope with complicated object details at diverse scales in HSI. In this article, we propose a deep multiscale pyramid network enhanced with spatial–spectral residual attention (DMPs^{2}raN) for HSI change detection, which has strong capability to mine multilevel and multiscale spatial–spectral features, improving the performance in complex changed regions. There are two key characteristics: 1) the multiscale spatial–spectral features are extracted by the multiscale pyramid convolution and enhanced by spatial–spectral residual attention module (S^{2}RAM) of each scale and 2) the multilevel features are obtained by aggregating the multiscale features level by level. As a result of this design, the proposed DMPs^{2}raN learns more discriminative features with both strong semantic information and rich spatial–spectral information. Experiments carried out on three datasets demonstrate the competitive performance of the proposed method in both qualitative and quantitative analyses.**

***Index Terms*— Change detection, convolutional neural network (CNN), hyperspectral image (HIS), multiscale feature extraction, spatial–spectral residual attention.**

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I. Introduction

AN OVERALL understanding of changes on the Earth is becoming increasingly important for better managing natural resources and preventing disasters. Change detection is a valuable tool to identify timely changes in land cover by analyzing bitemporal images over the same geographical area. With the increasing popularization and application of remote sensing, hyperspectral images (HSIs) have become one of the most commonly used data sources for Earth observation. Compared with multispectral images and synthetic aperture radar (SAR) images, HSIs provide a detailed spectral sampling of target ground objects over a wide range of electromagnetic spectra, which reflects the detailed composition with more spectral features [3], [4]. Therefore, the rich information from HSIs can effectively improve the performance of change detection. In recent years, HSI change detection has been developed and widely used in many important areas, including, but not limited to, natural disaster monitoring [5], urban landscape mapping [6], [7], and agricultural surveys.

Many change detection methods have been proposed to meet the requirements in various applications over the last few decades and can be classified into the following three categories: image algebra, image transformation, and classification-based algorithms.

A classic change detection approach based on image algebra is the change vector analysis (CVA) [8], which generates a difference image by calculating the Euclidean distance between pixels and then clusters the difference image to acquire the final change result. Bovolo *et al.* [9] further proposed a modified CVA model, named the compressed change vector analysis method (C^{2}VA), which provides a 2-D change representation in the polar domain to address the multiple change detection problem. However, the CVA-based methods, which measure differences by pixelwise comparison, are noise-sensitive and cannot take full account of spatial information.

Methods based on image transformation have the ability to transform HSIs to other feature spaces and highlight the changed regions in new spaces. The principal component analysis (PCA) is one of the most popular subspace learning algorithms with dimensionality reduction ability [10], [11]. However, PCA relies heavily on the statistical properties of the image and can be easily affected by unbalanced data. Multivariate alteration detection (MAD) proposed in [12] is

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designed to exact change objects by applying the canonical correlation analysis. The extended version, termed iterative reweighed MAD (IR-MAD) [13], is developed to assign a larger weight to unchanged features. Wu *et al.* [14] proposed a change detection method based on slow feature analysis (SFA), which can extract the most invariant component and transfers the input data to a new space, where unchanged pixels are suppressed, and the changed pixels can be better separated. However, the transformation-based methods still ignore the inner relationship between bitemporal bands.

Classification-based methods, including postclassification comparison [15], [16] and compound classification [17], can provide from-to information of land cover changes. Postclassification methods generate individual class maps of bitemporal images and compare the land-cover classification results between pixels. Certainly, the pixel misclassification in one image will lead to errors in the change detection result, no matter the corresponding pixel is classified correctly or not [18]. Moreover, the classification-based methods usually ignore the correlation between the bitemporal images and lead to unsatisfactory results.

The aforementioned methods only extract the handcrafted feature representations, limiting their performance in representing complex high-level change information. Recently, deep learning (DL) has shown impressive performances in computer vision and remote sensing [19]–[21]. Compared to traditional methods, DL models are capable of learning different features from different layers and handling the high dimensionality problems. Hence, a number of change detection approaches based on DL are developed. For change detection in optical aerial images, Liu *et al.* [22] introduced a deep depthwise separable convolutional network, which exploited U-Net to directly separate change and unchanged regions. Chen *et al.* [23] presented a multiscale feature convolution unit to deal with abundant spatial information provided by very high resolution (VHR) image, and two novel deep Siamese convolutional networks are developed for supervised and unsupervised change detection respectively. Li *et al.* [24] proposed a novel fully convolutional network framework, which is trained by an unsupervised noise modeling method for HSI change detection. A general end-to-end 2-D CNNs framework (GETNET) was adopted by Wang *et al.* [25], which introduced a mixed-affinity matrix to provide abundant cross-channel gradient information. Since the change detection task involves sequence data, the recurrent neural network (RNN) is introduced to model the temporal dependence in bitemporal images. Mou *et al.* [26] proposed a novel recurrent convolutional neural network (ReCNN) architecture for change detection in multispectral images, in which the spatial–spectral-temporal features are learned in an end-to-end manner. Bilinear convolutional neural networks (BCNNs) [27] were proposed by Lin *et al.* to learn spectral–spatial representations and obtain the combined bilinear features by linear outer product operation. Dual-branch difference amplification GCN (D^2AGCN) [28] first deployed the graph convolution network (GCN) with limited training samples to HSI change detection, which effectively extracts sufficient features and highlights the difference information.

The methods mentioned earlier do not take the abundant information in HSIs into consideration. Taking the advantage of the detailed spectral sampling, HSIs with high spectral resolution can identify minor spectral variations [29]. Compared with change detection from multispectral imagery, HSI change detection can identify finer change regions since more discriminate spatial–spectral information is provided by HSIs. However, most of the existing methods with a fixed kernel only take single-scale information into consideration, which is not powerful enough in feature representation. Therefore, for bitemporal HSIs containing complex changes, how to make full use of the multiscale spatial–spectral features is critical for improving the performance of HSI change detection.

To address the limitations of previous works, we propose a deep multiscale pyramid network enhanced with spatial–spectral residual attention (DMPs^{2}raN) to make full use of the multiscale features and aggregate them level by level to learn more discriminative information. In each level, the multiscale spatial–spectral features of HSIs are extracted by the multiscale pyramid convolution to cope with changed regions at diverse scales. The spatial–spectral features of each scale are enhanced with the spatial–spectral residual attention module (S^2RAM) to highlight the significant information. The representation ability of the model can be significantly improved, especially in the changed regions with detailed information. Since the network has relatively small receptive fields in shallow layers, the low-level features are suitable to capture detailed information. As the network goes deeper, the low-level features are integrated layer-by-layer to obtain the high-level semantic information, which is more abstract and robust. Therefore, we deepen the network to extract high-level semantic features that are obtained by aggregating the information within a larger receptive field. The low-level detail features and high-level semantic features are aggregated to make full use of the information in HSIs. In this article, the main contributions are listed as follows.

1) We propose an end-to-end network named DMPs^{2}raN to fully explore the multiscale and multilevel features, which has strong representation capability to deal with the diverse changes in HSIs.
2) The multiscale features are aggregated level by level, and both low- and high-level features are taken into account to capture rich spatial details and semantic information, improving the performance of the network.
3) The spatial–spectral residual attention module (S^2RAM) is designed to enhance the spatial–spectral feature of each scale, making the network pay more attention to significant information.

The remainder of this article is organized as follows. Section II details the proposed change detection framework. The experimental results and comparisons are presented and analyzed in Section III. Finally, the conclusion is drawn in Section IV.

II. Proposed Method

As shown in Fig. 1, the DMPs^{2}raN is designed to fully utilize the abundant information in HSI to identify change regions at different scales. The multiscale pyramid convolution

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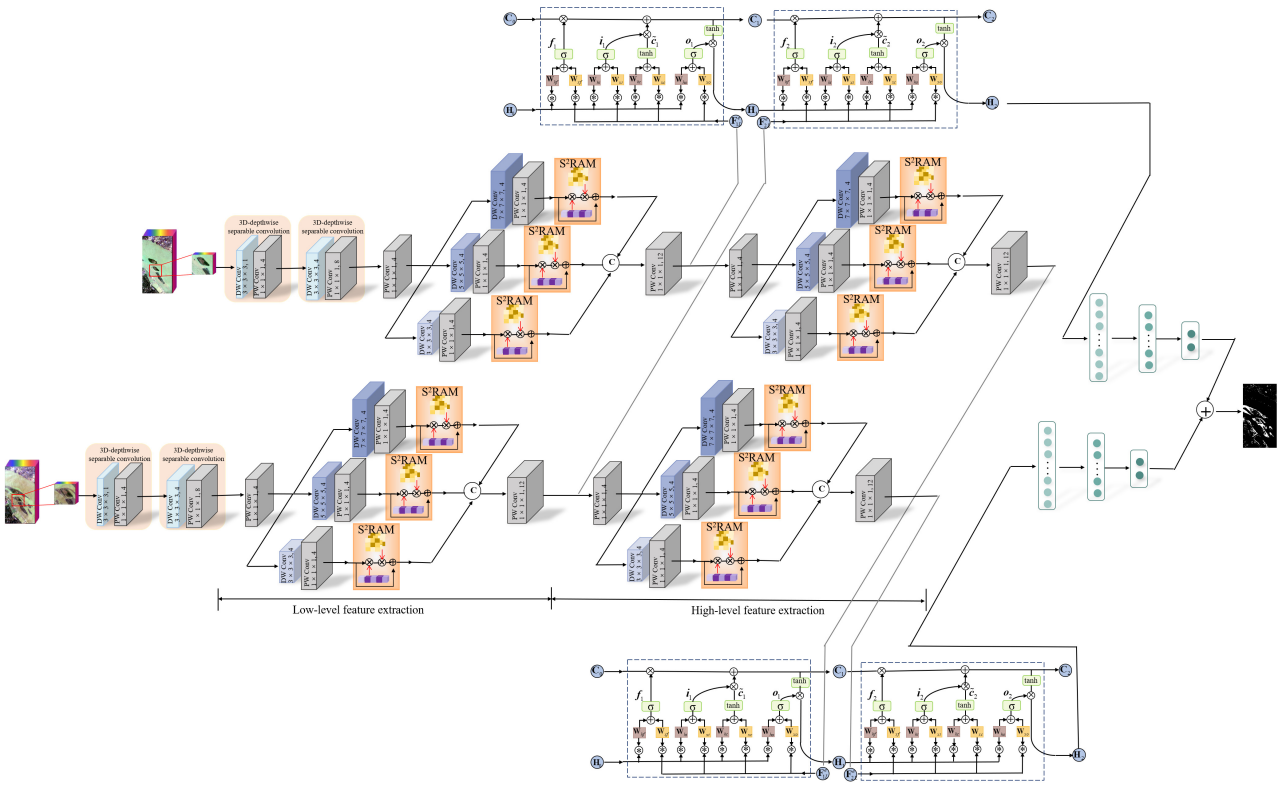


Fig. 1. Architecture of the proposed DMPs^{2}raN, which makes full use of the abundant information for HSI change detection.

is adopted to extract spatial–spectral features at different scales. For each scale, the S^{2}RAM is utilized to focus on the information related to the task. In order to further improve representation capability, high-level features with more abstract and robust information are utilized to assist with low-level spatial–spectral features.

In this section, the proposed HSI change detection method is described in detail. We first illustrate the baseline architectures and lead to the focus of this article on this basis. The loss function is also discussed, which makes the network end-to-end trainable.

A. Baselines

1) Baseline1-Multiscale Pyramid Feature Extraction: The change detection task involves two HSIs taken for the same location at different times. Thus, the existing networks have a symmetric structure with two branches, which ensures that deep features of the two images are extracted in the same way. The changed regions can be detected by calculating the difference between the extracted features. Given two HSIs $\mathbf{X}_1$ and $\mathbf{X}_2$, the extracted spatial–spectral features by the two-branch network can be universally formulated as

$$\begin{cases} \mathbf{F}_1 = f_1(\mathbf{X}_1) = C_1^l\left(\cdots C_1^i\left(\cdots C_1^1(\mathbf{X}_1)\right)\right) \\ \mathbf{F}_2 = f_2(\mathbf{X}_2) = C_2^l\left(\cdots C_2^i\left(\cdots C_2^1(\mathbf{X}_2)\right)\right) \end{cases} \tag{1}$$

where f_1 and f_2 are the feature extraction progress of two branches, $\mathbf{F}_1$ and $\mathbf{F}_2$ denote the output feature maps obtained by convolutional layers, l is the number of convolutional layers, and C_1^i and C_2^i are the ith convolutional layer of f_1 and f_2, respectively.

Most of the existing two-branch models adopt convolutional layers with a kernel size of 3×3 to extract features of the two images. Considering the fine spectral signatures provided by hyperspectral sensors, the HSIs provide abundant information with more details. The limited receptive field of 3×3 kernel size may be incompetent to cope with the multiscale spatial–spectral information in HSIs. Therefore, we adopt multiscale convolution kernels to extract complicated spatial–spectral information, improving the capability to detect subtle changes.

As illustrated in Fig. 2(a), the proposed multiscale pyramid convolution module in Baseline1 introduces 3-D convolution layers with different kernel sizes to extract multiscale features in parallel. The kernel with a small receptive field is exploited to effectively capture image details. Meanwhile, since the changed regions usually have various sizes, an increasing kernel size is capable of extracting sufficient contextual information from the image. The 3-D convolution can investigate the spectral correlation among adjacent bands and the spatial information among adjacent pixels simultaneously. Therefore, the spatial–spectral feature of HSI can be fully extracted by 3-D convolution. However, the number of parameters increases exponentially, which leads to high computational cost and limits the speed of change detection. Inspired by the bottleneck structure, we adopt 3-D pointwise convolution to reduce the dimension of feature maps and filter the low-dimension features with lightweight multiscale 3-D depthwise convolutions.

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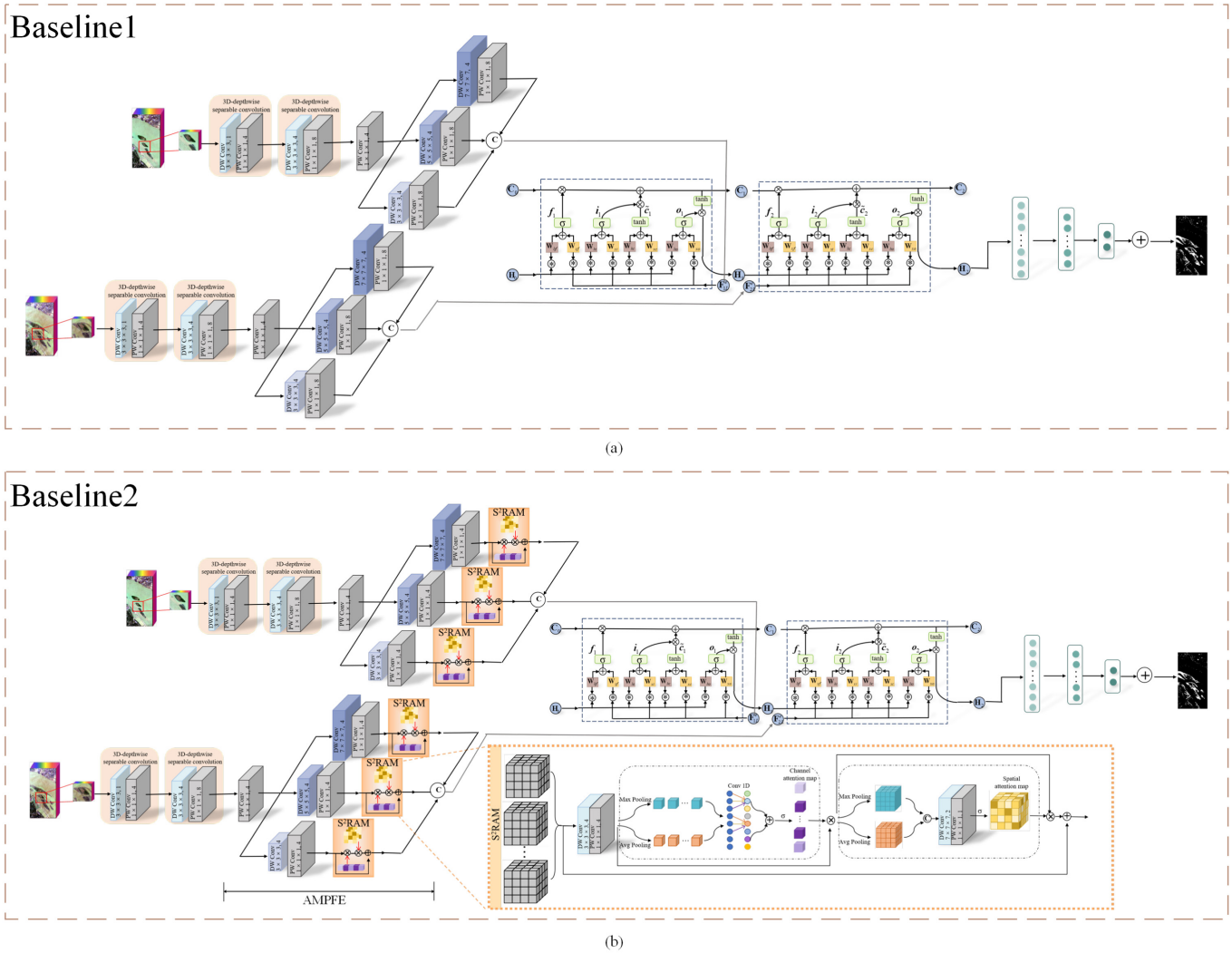


Fig. 2. Architectures of two baselines for HSIs' change detection. (a) Baseline1 with multiscale pyramid convolution to extract multiscale spatial–spectral features. (b) Baseline2 with the proposed AMPFE block to obtain a more powerful feature representation.

Given the input $\mathbf{X}_t$ $(t = 1, 2)$, the 3-D depthwise separable convolutional layers [30], which replaces the standard 3-D convolution layers with a 3-D depthwise convolution and a 3-D pointwise convolution, are adopted to extract the spatial–spectral features $\mathbf{I}_t$ of the input. The $3 \times 3 \times 3$ depthwise convolution applies separate convolutional filters per input channel, and a pointwise convolution ($1 \times 1 \times 1$ convolution) carries out a linear combination of the output of the depthwise convolution. The spatial–spectral features $\mathbf{I}_t$ of the input is formulated as

$$\mathbf{I}_t = \mathrm{DSC}_t^l\left(\cdots \mathrm{DSC}_t^i\left(\cdots \mathrm{DSC}_t^1(\mathbf{X}_t)\right)\right) \tag{2}$$

where DSC_t^i is the ith 3-D depthwise separable convolutional layer. Then, the multiscale features $\mathbf{O}_t^k$ of the two branch generated by convolutional layers with different kernel sizes can be represented as

$$\mathbf{O}_t^k = f_{\mathrm{LConv3D}}^k(\mathbf{I}_t) = f_{PW}\left(f_{DW}^k(f_{PW}(\mathbf{I}_t))\right), \quad k = 1, 2, 3 \tag{3}$$

where $f_{\mathrm{LConv3D}}^k(\cdot)$ denotes the lightweight 3-D convolution operation with the kth type of filter, and $f_{PW}(\cdot)$ and $f_{DW}^k(\cdot)$ are the pointwise convolution layer with the kernel size of $1 \times 1 \times 1$ and the depthwise convolution layer with the kth type of 3-D filter (we use the 3-D filter of size $3 \times 3 \times 3$, $5 \times 5 \times 5$, and $7 \times 7 \times 7$ in this module). The extracted multiscale feature maps $\mathbf{O}_t^k$ of each branch are concatenated in parallel, which can be expressed as

$$\mathbf{F}_t = \mathrm{Concat}\left(\mathbf{O}_t^1, \mathbf{O}_t^2, \mathbf{O}_t^3\right) \tag{4}$$

where $\mathrm{Concat}(\cdot)$ denotes the concatenate operation and $\mathbf{F}_t$ $(t = 1, 2)$ is the output feature map of each branch in Baseline1.

2) Baseline2-Attention-Enhanced Multiscale Pyramid Feature Extraction: In Baseline1, the multiscale pyramid convolution is applied to extract features at diverse scales in HSIs. However, multiscale features contain redundant information, and the significant features may be influenced by useless information. Recently, the attention mechanism has received increasing attention in the field of computer vision [31], [32]. Inspired by the human visual system, the attention mechanism can allocate the limited resource to focus on the most valuable and informative stimulus. In order to refine significant features, the spatial–spectral residual

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attention module (S^2RAM) is adopted at each scale on the basis of Baseline1. As shown in Fig. 2(b), we construct Baseline2 to learn attention-enhanced multiscale feature representations, which improves the performance of change detection.

The detailed architecture of S^2RAM is illustrated in Fig. 2(b). Different information should be treated differently in the feature map of different scales, so the S^2RAM is embedded to emphasize significant information of each scale. The CNN models have limited ability to distinguish different kinds of features and treat them equally. The channel attention module is adopted to focus on "what" is meaningful in the input, improving the representation ability of the model. Instead of adopting fully connected (Fc) layers that are widely used to capture nonlinear cross-channel interaction, we use a 1-D convolutional layer to avoid dimension reduction, which may lead to side effects [33]. Both global average pooling and global maximum pooling are adapted simultaneously to aggregate spatial information of the feature to generate two different spatial information descriptors. Then, the two descriptors are fed into a 1-D convolutional layer with kernel size of 3 to capture local cross-channel interaction, and the filter weights are shared. The output features are merged with elementwise summation and normalized to the range from 0 to 1 with a sigmoid activation to obtain the desired channel attention map $\mathbf{C}_t^k$, which can be formulated as

$$\mathbf{C}_t^k = \sigma\left(f_{\text{Conv1D}}\left(\text{AvgPool}\left(\text{DSC}_t\left(\mathbf{O}_t^k\right)\right)\right) + f_{\text{Conv1D}}\left(\text{MaxPool}\left(\text{DSC}_t\left(\mathbf{O}_t^k\right)\right)\right)\right) \quad (5)$$

where f_{Conv1D} is the 1-D convolutional layer of kernel size 3, DSC_t represents the 3-D depthwise separable convolutional layer, σ denotes the sigmoid activation function, and AvgPool$(\cdot)$ and MaxPool$(\cdot)$ are the average pooling and maximum pooling operations, respectively.

Most of the existing methods are able to identify the main changed areas, while the details such as object boundaries and scattered changed regions are failed to be detected. The accuracy of change detection highly depends on the performance in complicated regions with more details. However, the multilayer convolution operation may lead to the loss of information, resulting in the elimination of the features of the scattered changed regions with only a few pixels. Therefore, the spatial attention is employed in the S^2RAM to focus on "where" contains information related to the task of the feature of each scale. The pixels that belong to different category from the center pixel are assigned smaller weights to reduce the interference of neighboring pixels of the patch in the region with complicated details. The input feature maps are aggregated by average pooling and maximum pooling along with the channel dimension to highlight informative regions. The outputs are concatenated, and a 3-D depthwise separable convolution layer of kernel size $7 \times 7 \times 7$ is applied to produce the spatial attention map. The spatial attention map $\mathbf{S}_t^k$ can be computed as

$$\mathbf{S}_t^k = \sigma\left(\text{DSC}_t\left(\text{Concat}\left(\text{AvgPool}\left(\mathbf{C}_t^k \otimes \text{DSC}_t\left(\mathbf{O}_t^k\right)\right), \text{MaxPool}\left(\mathbf{C}_t^k \otimes \text{DSC}_t\left(\mathbf{O}_t^k\right)\right)\right)\right)\right) \quad (6)$$

where $\otimes$ represents elementwise multiplication.

Particularly, we combine the channel attention and spatial attention into a residual block to gain more informative feature. Therefore, the extracted multiscale feature maps $\mathbf{O}_t^k$ are enhanced by the proposed S^2RAM, which can be expressed as

$$\mathbf{O}_t^{ek} = \text{DSC}_t(\mathbf{O}_t^k) \otimes \mathbf{C}_t^k \otimes \mathbf{S}_t^k + \mathbf{O}_t^k \quad (7)$$

where $\mathbf{O}_t^{ek}$ is the attention-enhanced feature map obtained by the S^2RAM.

With the ES^2RA module embedded, the proposed attention-enhanced multiscale pyramid feature extraction (AMPFE) block possesses a powerful ability to extract and enhance multiscale features. Given the input feature map $\mathbf{I}_t$, the multiscale features extracted by the multiscale pyramid convolution in Baseline1 are enhanced by S^2RAM to highlight the task-related information of each scale. The attention-enhanced multiscale feature map $\mathbf{O}_t^{ek}$ in Baseline2 can also be expressed as

$$\mathbf{O}_t^{ek} = f_{\text{S}^2\text{RAM}}\left(\mathbf{O}_t^k\right) = f_{\text{S}^2\text{RAM}}\left(f_{\text{LConv3D}}^k(\mathbf{I}_t)\right), \quad k = 1, 2, 3 \quad (8)$$

where $f_{\text{S}^2\text{RAM}}(\cdot)$ represents the function of S^2RAM. Finally, the attention-enhanced multiscale feature maps of each branch $\mathbf{O}_t^{ek}$ are concatenated to obtain the output of AMPFE block $\mathbf{F}_t^e$ in Baseline2, which can be calculated by

$$\mathbf{F}_t^e = \text{Concat}(\mathbf{O}_t^{e1}, \mathbf{O}_t^{e2}, \mathbf{O}_t^{e3}). \quad (9)$$

B. Proposed Deep Multiscale Pyramid Network Enhanced With Spatial–Spectral Residual Attention

As mentioned in Section A, we apply the multiscale pyramid convolution in Baseline1 to learn spatial–spectral features at different scales, which is effective to cope with the complicated changed regions. In order to further improve feature representation capability, the AMPFE block containing S^2RAM at each scale is proposed to obtain attention-enhanced features by highlighting the influence of useful information and suppressing the insignificant region in Baseline2. However, low-level features with fine details may be difficult to capture the semantic information. Based on the architectures of Baseline1 and Baseline2, we deepen the network in the proposed deep multiscale attention-enhanced pyramid network to extract high-level information. The details of the proposed DMPs^{2}raN, as shown in Fig. 1, are given as follows.

The hyperspectral patches $\mathbf{X}_1$ and $\mathbf{X}_2$ of size $9 \times 9 \times C$ are fed into the network in parallel, where C denotes the number of channels. We follow the architecture of AMPFE block described in Section A for effective multiscale feature extraction. The proposed AMPFE block captures multiscale information to cope with changed regions at diverse scales and adopts S^2RAM to emphasize informative region of feature map at each scale. Given $\mathbf{I}_t$ as the input feature map obtained by $3 \times 3 \times 3$ depthwise separable convolutional layers, the low-level attention-enhanced multiscale feature $\mathbf{O}_{tL}^{ek}$ can be expressed as

$$\mathbf{O}_{tL}^{ek} = f_{\text{S}^2\text{RAM}}\left(f_{\text{LConv3D}}^k(\mathbf{I}_t)\right), \quad k = 1, 2, 3. \quad (10)$$

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Furthermore, a feature fusion layer is employed to fuse low-level features at different scales after being concatenated in the AMPFE block. The multiscale feature fusion process can be formulated as

$$\mathbf{F}_{tL}^{e} = f_{PW}(\text{Concat}(\mathbf{O}_{tL}^{e1}, \mathbf{O}_{tL}^{e2}, \mathbf{O}_{tL}^{e3})) \tag{11}$$

where $\mathbf{F}_{tL}^{e}$ denotes the fused low-level feature, which is the output of the $1 \times 1 \times 1$ pointwise convolution layer.

Since the HSI contains complex ground object conditions, the low-level feature $\mathbf{F}_{tL}^{e}$ may not powerful enough to mine discriminative information. To address this issue, we deepen the network to obtain more abundant features. Due to the representation capability of convolution layers, the features capture high-level semantic information with the layer going deeper. Similar to the low-level feature extraction, the AMPFE block followed by a feature fusion layer is utilized to obtain the high-level feature of each branch. The high-level multiscale features $\mathbf{O}_{tH}^{ek}$ can be formulated as

$$\mathbf{O}_{tH}^{ek} = f_{\text{S}^2\text{RAM}}\left(f_{\text{LConv3D}}^{k}\left(\mathbf{F}_{tL}^{e}\right)\right), \quad k = 1, 2, 3. \tag{12}$$

The extracted high-level features are fused by the $1 \times 1 \times 1$ pointwise convolution layer, which can be expressed as

$$\mathbf{F}_{tH}^{e} = f_{PW}\left(\text{Concat}\left(\mathbf{O}_{tH}^{e1}, \mathbf{O}_{tH}^{e2}, \mathbf{O}_{tH}^{e3}\right)\right) \tag{13}$$

where $\mathbf{F}_{\text{tH}}^{e}$ denotes the fused high-level feature.

The aforementioned CNN-based network automatically extracts multiscale spatial–spectral features without manual intervention. However, the network treats the inputs independently, ignoring the temporal correlation between them. Since the input images are acquired in the same area during a period of time, it is reasonable to assume that there exists temporal connection between bitemporal images. RNN is used to model the temporal features of the bitemporal images [34], [35]. The long short-term memory (LSTM) is an extensive version of RNN and was initially proposed by Hochreiter *et al.* [36]. Compared with traditional RNN, LSTM overcomes the vanishing gradient problem and performs better in more challenging tasks involving sequential time series. Convolutional LSTM (ConvLSTM) [37] further replaces the full connection layers in LSTM with convolutional layers, which receives the 2-D features as input without vectorized, preventing the loss of spatial correlation information. Containing three key gates, the input gate, the output gate, and the forget gate, the ConvLSTM can perform information protection and control. The calculation formulas of the three gates in each unit can be expressed as

$$\begin{cases} \boldsymbol{i}_t = \sigma\left(\mathbf{W}_{xi} * \mathbf{F}_{t.}^{e} + \mathbf{W}_{hi} * \mathbf{H}_{t-1} + \boldsymbol{b}_i\right) \\ \boldsymbol{f}_t = \sigma\left(\mathbf{W}_{xf} * \mathbf{F}_{t.}^{e} + \mathbf{W}_{hf} * \mathbf{H}_{t-1} + \boldsymbol{b}_f\right) \\ \boldsymbol{o}_t = \sigma\left(\mathbf{W}_{xo} * \mathbf{F}_{t.}^{e} + \mathbf{W}_{ho} * \mathbf{H}_{t-1} + \boldsymbol{b}_o\right) \end{cases} \tag{14}$$

where $\mathbf{F}_{t.}^{e}$ denotes the low-/high-level spatial–spectral feature $\mathbf{F}_{tL}^{e}$ or $\mathbf{F}_{tH}^{e}$; $\boldsymbol{i}_t$, $\boldsymbol{f}_t$, and $\boldsymbol{o}_t$ are all 3-D tensors denoting the input gate, the forget gate, and the output gate, respectively; $\mathbf{W}_{\cdot i}$, $\mathbf{W}_{\cdot f}$, and $\mathbf{W}_{\cdot o}$ are the weight matrix of the input gate, the forget gate, and the output gate; $\boldsymbol{b}_i$, $\boldsymbol{b}_f$, and $\boldsymbol{b}_o$ are the bias of these three gates; the activation function $\sigma(\cdot)$ is a sigmoid function; and $*$ denotes the convolution operation.

The memory cell $\mathbf{C}_t$ at time step t is updated by partially adding the candidate cell value $\tilde{\mathbf{C}}_t$. The update of the output hidden state $\mathbf{H}_t$ is implemented as

$$\tilde{\mathbf{C}}_t = \tanh\left(\mathbf{W}_{xc} * \mathbf{F}_{t.}^{e} + \mathbf{W}_{hc} * \mathbf{H}_{t-1} + \boldsymbol{b}_c\right) \tag{15}$$

$$\mathbf{C}_t = \boldsymbol{f}_t \circ \mathbf{C}_{t-1} + \boldsymbol{i}_t \circ \tilde{\mathbf{C}}_t \tag{16}$$

$$\mathbf{H}_t = \boldsymbol{o}_t \circ \tanh(\mathbf{C}_t) \tag{17}$$

where $\mathbf{H}_{t-1}$ is the hidden state at previous time step $t-1$, $\mathbf{W}_{\cdot c}$ and $\boldsymbol{b}_c$ are the weight matrix and bias of the memory cell, $\tanh(\cdot)$ is the hyperbolic tangent function, and $\circ$ represents the Hadamard product.

Both low- and high-level features are fed as the input of the ConvLSTM network in parallel to model temporal dependence. For the change detection task, the last hidden states of different levels $\mathbf{H}_{2L}$ and $\mathbf{H}_{2H}$ are taken as the output of ConvLSTM. After being flattened as the input of Fc layers, the results of different levels are fused to obtain the binary classification result, which can be formulated as

$$\mathbf{CM} = \text{softmax}(\text{Fc}(\boldsymbol{h}_L) + \text{Fc}(\boldsymbol{h}_H)) \tag{18}$$

where $\mathbf{CM}$ denotes the change map, $\text{Fc}(\cdot)$ represents the Fc layers, and $\text{softmax}(\cdot)$ is the softmax function for binary change detection. $\boldsymbol{h}_L$ and $\boldsymbol{h}_H$ are the inputs of Fc layers obtained by flattening the output hidden states $\mathbf{H}_{2L}$ and $\mathbf{H}_{2H}$ of different levels.

C. Loss Function

To optimize the proposed model, an objective function is needed. The change detection task can be treated as a binary classification problem. Therefore, cross-entropy, which is commonly used for classification, is adopted as the loss function of DMPs^{2}raN to measure the similarity between the ground truth and the predicted change map. The binary cross-entropy can be expressed as

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{n}\sum_{i=1}^{n}(y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)) \tag{19}$$

where n denotes the number of samples, y_i is the label of ground truth, whose value is 0 or 1, and $\hat{y}_i$ represents the predicted probability value, which is calculated by the softmax function.

III. Experiment Results and Analysis

A. Data Description

The Yancheng dataset, as shown in Fig. 3(a) and (b), covers farmland for the city of Yancheng, Jiangsu Province, China. The two HSIs are obtained by the Earth Observing-1 (EO-1) Hyperion sensors on May 3, 2006, and April 23, 2007, respectively. The EO-1 Hyperion sensor offers 242 spectral bands ranging from 0.4 to 2.5 μm and a spatial resolution of 30 m [38], [39]. Each image has a spatial size of 420×140 pixels and contains 154 spectral bands for change detection after removing noisy bands. It includes various scenes, such as trees, soil, river, and agricultural field, among which the major change areas are land cover and rivers.

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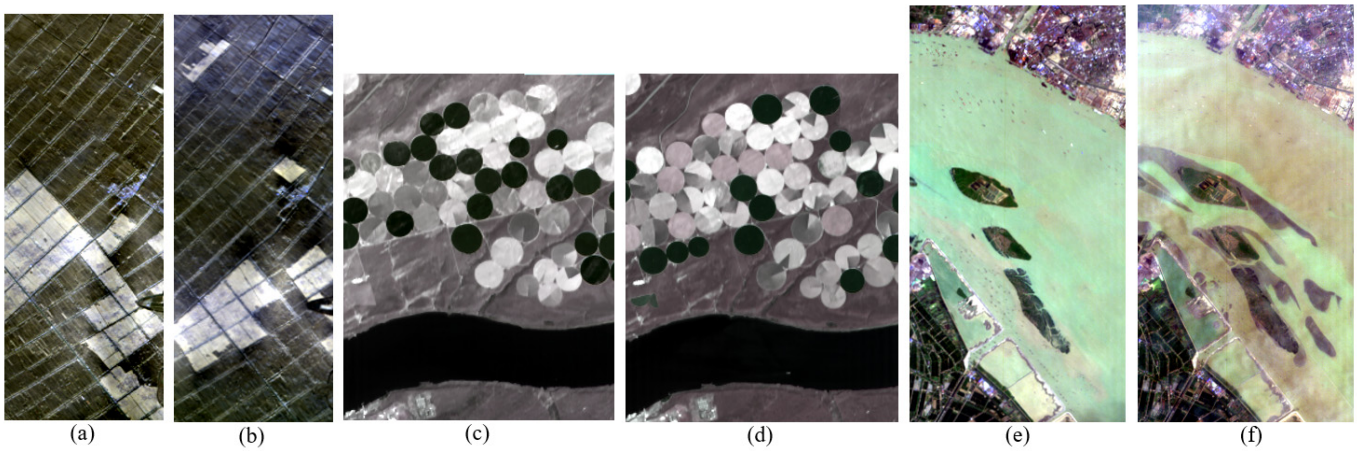


Fig. 3. Pseudocolor images of the datasets in this article. (a) Yancheng image acquired on May 3, 2006. (b) Yancheng image acquired on April 23, 2007. (c) Hermiston image acquired on May 1, 2004. (d) Hermiston image acquired on May 8, 2007. (e) Jiangsu image acquired on May 3, 2013. (f) Jiangsu image acquired on December 31, 2013.

The Hermiston dataset containing two images is shown in Fig. 3(c) and (d), which depicts an irrigated agricultural field. The images were acquired in Hermiston city in Umatilla County, Oregon, OR, USA, on May 1, 2004, and May 8, 2007, respectively. The size of each image is 307 × 241 pixels with 154 spectral bands after noise elimination, providing a spectral resolution of 10 nm. The changes on the Hermiston dataset are related to river and land cover.

The Jiangsu dataset, as shown in Fig. 3(e) and (f), was collected on May 3, 2013, and December 31, 2013 [25], respectively. The data consist of two images, covering Jiangsu Province, China. For this dataset, the size of each image is 463 × 241 pixels with the spatial resolution 30 m, and there are 198 bands selected for this experiment after discarding the noisy bands. The vanishment of substance in the river is the main change type in this dataset.

Change detection can be regarded as a binary classification problem, in which pixels are divided into 'changed' and 'unchanged' classes. Since the proportions of these two classes are different, we first calculate the proportions of changed and unchanged pixels in each dataset. Next, we randomly select 20% pixels as training set according to the ratio, and the remaining 80% pixels are used for testing. In order to eliminate the influence of randomly selected samples, CNN-based methods run four times per dataset, and we calculate the average value and standard deviation as the final results.

B. Evaluation Measures

For quantitative assessment, the overall accuracy (OA) and the kappa coefficient (KC) are adopted to evaluate the change maps. Specifically, OA shows the ratio of the number of correctly classified pixels to the number of the total test sample pixels. The larger the value of OA is, the better the change detection result of the method obtains. The OA is calculated as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{20}$$

TABLE I
CONFUSION MATRIX

Confusion Matrix		Predicted	
		Change	Unchange
Actual	Change	TP	FN
	Unchange	FP	TN

where TP means true positives, TN means true negatives, FP means false positives, and FN means false negatives. The confusion matrix is illustrated in Table I, which provides more details of these four quantities.

As for KC [40], it is an index that indicates the accuracy of classification. It represents the degree of similarity between the experimental result and the ground-truth map in change detection. The higher the KC value is, the better the result of change detection is. The KC is defined as

$$\mathrm{KC} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{21}$$

where

$$\mathrm{PRE} = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} + \frac{(\mathrm{FN}+\mathrm{TN})(\mathrm{FP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2}. \tag{22}$$

C. Experimental Results

To verify the effectiveness of our proposed method, several state-of-the-art methods, such as CVA [8], PCA [10], IR-MAD [13], support vector machines (SVMs) [41], ReCNN [26], BCNNs [27], and D^2AGCN [28], were considered for comparison. As a classical unsupervised change detection method, CVA makes full use of the information of each available spectral channel. The PCA method preserves the key information of the image by dimensionality reduction, reducing the computational complexity. IR-MAD is an important HSI change detection method, which emphasizes the changed regions by applying different weights to each pixel.

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TABLE II
QUANTITATIVE ASSESSMENT OF DMPes^{2}raN AND OTHER STATE-OF-THE-ART CHANGE DETECTION METHODS ON THREE DATASETS

Dataset	Yancheng		Hermiston		Jiangsu	
Index	OA(%)	KC	OA(%)	KC	OA(%)	KC
CVA	87.49	0.6998	92.00	0.7410	92.67	0.6575
PCA	88.92	0.7322	91.93	0.7366	95.37	0.7538
IR-MAD	89.74	0.7632	90.25	0.6723	93.82	0.6798
SVM	91.12	0.7943	95.76	0.8740	97.40	0.8292
D^{2}AGCN	96.45±0.21	0.9173±0.0034	95.78±0.15	0.8773±0.0027	96.32±0.18	0.7574±0.0058
BCNNs	97.51±0.16	0.9441±0.0016	96.13±0.12	0.8867±0.0034	96.13±0.12	0.7435±0.0064
ReCNN	97.47±0.12	0.9412±0.0028	95.93±0.09	0.8797±0.0025	95.97±0.25	0.7337±0.0073
DMPs^{2}raN	**97.87±0.26**	**0.9504±0.0062**	**96.96±0.14**	**0.9124±0.0042**	**97.80±0.08**	**0.8586±0.0041**

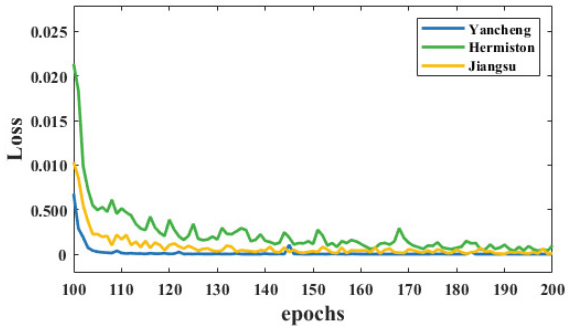


Fig. 4. Training losses of the proposed DMPs^{2}raN on three datasets.

The SVM method calculates a high-dimensional hyperplane to distinguish changed and unchanged parts of the image. With respect to the DL-based method, ReCNN, BCNNs, and D^{2}AGCN are chosen as competing methods for their excellent performance.

The network is trained based on the PyTorch on an NVIDIA GTX 2080Ti. We choose Adam [42] as the optimizer to train the network. The batch size is set to 32, and we train the network for 200 epochs. The initial learning rate of the network is 1e-4 and is divided by 10 every 100 epochs. To investigate the convergence of the proposed model, Fig. 4 shows the training curves of the DMPs^{2}raN on three datasets, respectively. From Fig. 4, it can be observed that the proposed DMPs^{2}raN has a small loss and converges rapidly to the fine performance during 110 epochs on all three datasets.

1) Experiments on the Yancheng Dataset: The visual comparison between the results of different methods on the Yancheng dataset is shown in Fig. 5. Changes are mainly about the size of farmland. It can be clearly observed that there exist many white noise points in the change map of the CVA method. Simple algebraic operations cannot generate an accurate difference image, resulting in poor performance. Similar to the result of CVA, the change map produced by PCA also missed many changed regions. The PCA method loses some useful information due to dimensionality reduction, which leads to the occurrence of misjudgment. Although the performance of IR-MAD is better than that of CVA and PCA, there are still many unchanged pixels that are misdetected as changed pixels. The SVM method failed to identify the main change areas. ReCNN shows poor performance on the Yancheng dataset, especially on the edge of farmland. The change maps generated by BCNNs and D^{2}AGCN are superior to the traditional methods with fewer noise points. It is obvious that the changed area generated by our proposed method is relatively smooth with fine edges and fewer noisy spots. The reason is that the proposed method makes full use of the spatial–spectral information of HSIs to improve the representation capability of complicated ground objects.

As reported in Table II with the best performance being highlighted in bold, all the unsupervised methods, including CVA, PCA, and IR-MAD, yield OA lower than 90%, indicating the low classification accuracy of these methods. The SVM method obtains better results than other traditional methods, providing OA of 91.12% and KC of 0.7943. The DL-based methods obtain higher values in terms of both OA and KC. The values of OA and KC of D^{2}AGCN are lower than ReCNN and BCNNs, indicating the influence of superpixel segmentation on the accuracy of change detection. By capturing both low- and high-level features, DMPs^{2}raN achieves OA of 98.18% and KC of 0.9576, respectively. The proposed DMPs^{2}raN outperforms all the competing methods with the highest OA and KC on the Yancheng dataset, which demonstrates its effectiveness.

2) Experiments on the Hermiston Dataset: As shown in Fig. 6, the major changes occurred in the area covered by farmland, and some areas, such as the edge of the river, also had some minor changes. It can be seen from Fig. 6(b) that many changed areas cannot be detected by the CVA method. The changes in many areas in the change maps of PCA and IR-MAD methods were ignored, such as the edge of the river and part of the farmland area. It can be seen from Fig. 6(e) that the performance of the SVM methods is relatively ideal, and the missed detection phenomenon is alleviated, but there are still some small changes that are not detected. The performance of ReCNN and D^{2}AGCN is very close to that of BCNNs, all of which generate change maps with more details. Since the Hermiston dataset contains more complicated changed regions, stronger representation capability is required to obtain better results. It is obvious that the method we proposed has earned the best results with the least missed detections or false detections, especially the complicated changes in the right of the image.

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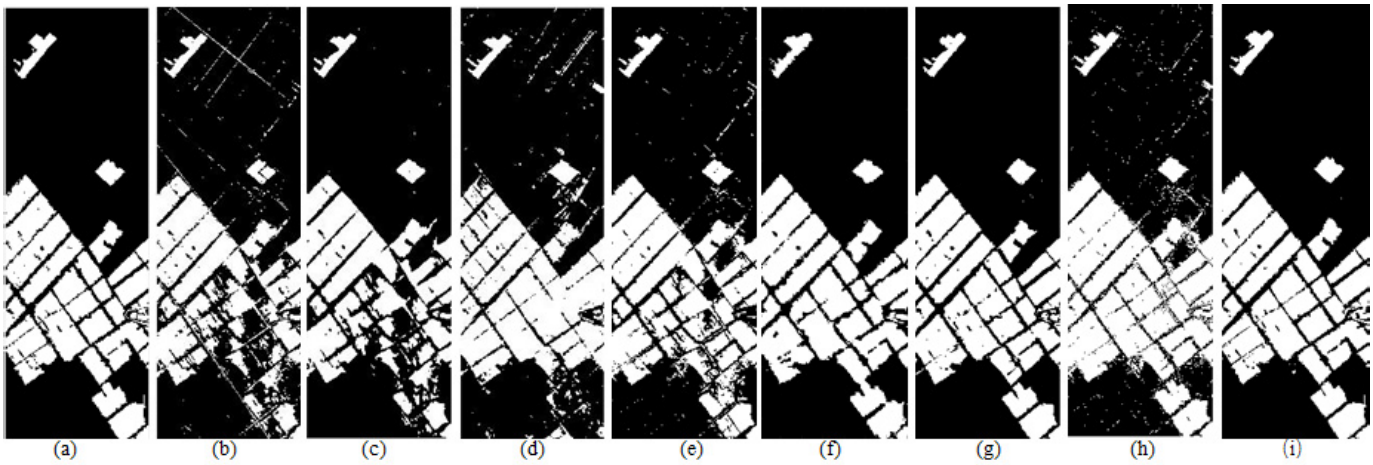


Fig. 5. Change maps of different methods on the Yancheng dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) D^2AGCN. (g) BCNNs. (h) ReCNN. (i) DMPs^{2}raN.

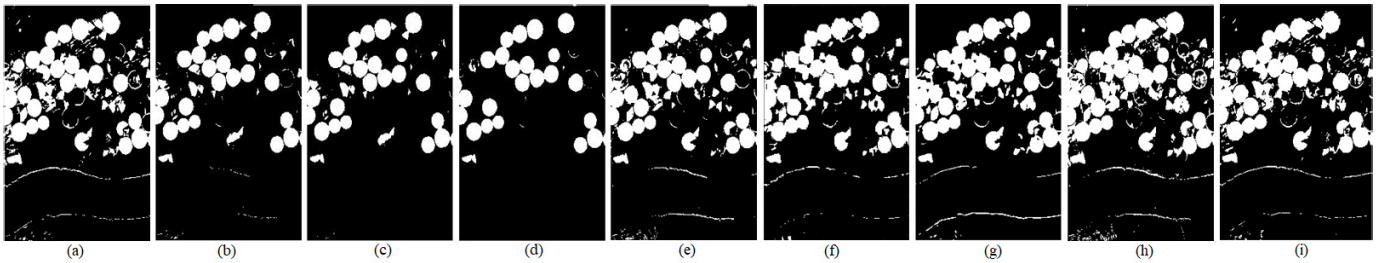


Fig. 6. Change maps of different methods on the Hermiston dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) D^2AGCN. (g) BCNNs. (h) ReCNN. (i) DMPs^{2}raN.

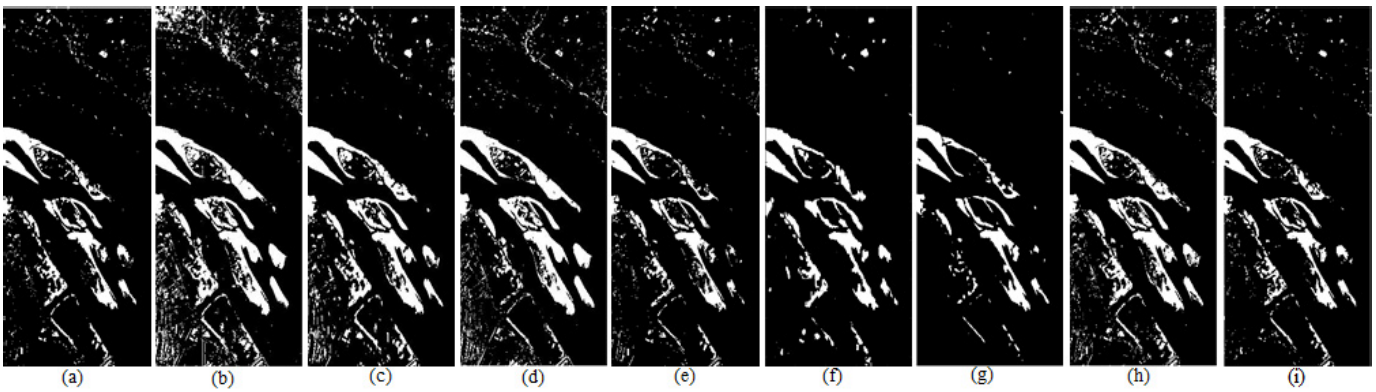


Fig. 7. Change maps of different methods on the Jiangsu dataset. (a) Ground-truth map. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) D^2AGCN. (g) BCNNs. (h) ReCNN. (i) DMPs^{2}raN.

It can be seen from Table II, IR-MAD achieves the worst result with OA of 90.25% and KC of 0.6723 on the Hermiston dataset, and the poor performance of the IR-MAD method indicates its insensitivity to complicated changes. In addition, the performance of SVM is better than the unsupervised methods, generating OA up to 95% and KC of over 0.87. Compared with the traditional methods, the DL-based methods show better performance with OA and KC more than 95.7% and 0.877, respectively. Our method obtains the highest values on both OA and KC. This is because we have fully extracted multilevel and multiscale spatial–spectral features, thereby further improving the accuracy of the HSI change detection method.

3) Experiments on the Jiangsu Dataset: As shown in Fig. 7, the most obvious change is the increase of material content in rivers, while there are some sporadic changes in other regions. It is obvious that the experimental results are different due to the different characteristics of each method. Specifically, since the CVA method detects the change areas by computing the Euclidean distance between each pixel, it is very sensitive to noise, which can be strongly proved by the noise white points in Fig. 7(b). Meanwhile, we can clearly see

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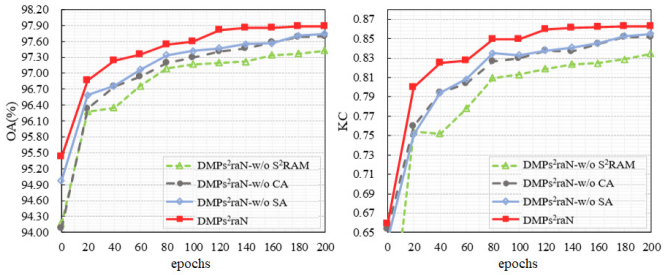


Fig. 8. Training curves of models with different configurations of attention on the Jiangsu dataset.

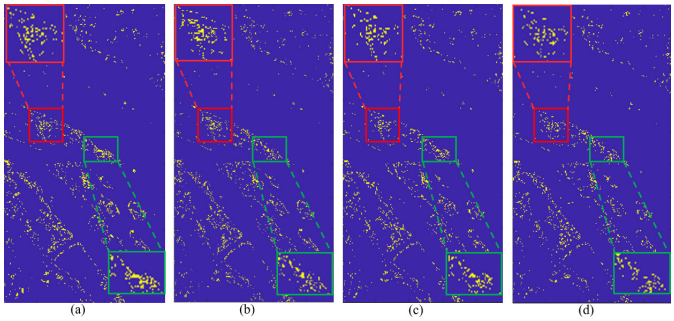


Fig. 9. Difference images between the change maps generated by different models and ground-truth map on the Jiangsu dataset. (a) DMPs^{2}raN-w/o S^{2}RAM. (b) DMPs^{2}raN-w/o CA. (c) DMPs^{2}raN-w/o SA. (d) DMPs^{2}raN.

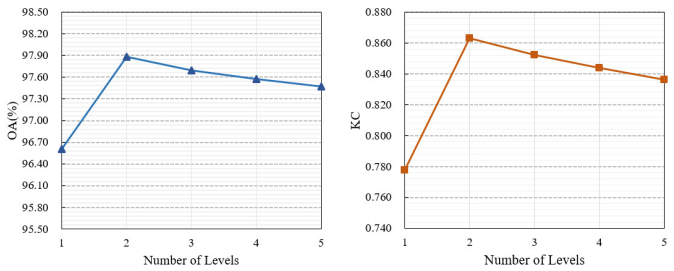


Fig. 10. Detection performance curves of DMPes^{2}raN under different network depth settings on the Jiangsu dataset.

that many unchanged areas in the lower left corner of the image are incorrectly classified as changed by the PCA and IR-MAD methods. Although the integrity of detected regions is improved, there still exist false alarms and missed targets in the result obtained by the SVM method. As can be seen in Fig. 7(g), BCNNs have poor performance on the Jiangsu dataset with massive missed targets. D^{2}AGCN achieves better results, which detects the complicated changes in the upper right corner. The change map generated by ReCNN is more satisfactory, but the false alarm rate is high. The result of the proposed DMPs^{2}raN method is shown in Fig. 7(i), which shows the best performance and can accurately detect obvious or subtle changes with almost no noise interference.

Table II also records the metric values obtained on the Jiangsu dataset. The values of OA and KC obtained by CVA and IR-MAD methods are obviously lower than other methods. The results obtained by the supervised methods, including SVM, ReCNN, BCNNs, and D^{2}AGCN, are relatively better, all generating OA more than 96% and KC up to 0.72. It is obvious that our proposed method owns the best experimental results with OA of 97.88% and KC of 0.8630, which shows that it can effectively learn image features for change detection. The reason is that the proposed DMPs^{2}raN extracts the multiscale features and aggregates them level by level to make full use of the information in HSIs.

D. Ablation Studies

1) Effectiveness of S^{2}RAM: We conduct ablation experiments on the Jiangsu dataset to verify the effectiveness of the proposed S^{2}RAM. To further distinguish the influence of channel attention and spatial attention, the experiments are also carried out on S^{2}RAM without channel attention or spatial attention. We construct four models with different configurations, i.e., DMPs^{2}raN-w/o S^{2}RAM, which removes the proposed S^{2}RAM module in DMPs^{2}raN, DMPs^{2}raN-w/o CA with only spatial attention preserved in ES^{2}RA, DMPs^{2}raN-w/o SA with channel attention preserved in S^{2}RAM, and DMPs^{2}raN. The results of these four models are shown in Fig. 8, and the parameters of each model are tuned to obtain their best performance. It can be seen that each evaluation index of the DMPs^{2}raN-w/o S^{2}RAM is significantly lower than that of the proposed DMPs^{2}raN, indicating the effectiveness of the proposed S^{2}RAM. Compared with DMPs^{2}raN-w/o S^{2}RAM, both DMPs^{2}raN-w/o SA and DMPs^{2}raN-w/o CA have a significant improvement. The OA and KC of DMPs^{2}raN-w/o SA are 97.74% and 0.8549, which corresponds to increases of 0.3% and 0.0205. The spatial attention in S^{2}RAM that highlights the task related information is able to improve the accuracy of change detection. The proposed S^{2}RAM combines both channel attention and spatial attention into residual block, achieving the best performance in terms of OA and KC.

For the qualitative analysis, we show the differences between the change maps generated by different models and ground-truth map on the Jiangsu dataset in Fig. 9, respectively. As shown in Fig. 9(a), DMPs^{2}raN-w/o S^{2}RAM fails to accurately identify the boundaries of the changed regions. The results of DMPs^{2}raN-w/o CA and DMPs^{2}raN-w/o SA are presented in Fig. 9(b) and (c), which performs better than that of DMPs^{2}raN-w/o S^{2}RAM. With the help of S^{2}RAM, the proposed DMPs^{2}raN achieves the best performance with fine boundaries and accurately detects the scatted changed regions (see the red and green rectangle regions). The experimental results demonstrate that the proposed S^{2}RAM can highlight the significant spatial and spectral features, improving the accuracy of HSIs' change detection.

2) Depth of the Network: Different layers of the network have different receptive fields, thus providing rich hierarchical features. The low-level feature is suitable to capture detailed information, such as some changed regions with only a few pixels. On the contrary, the high-level feature, which is generated by deeper layers with large receptive fields, is able to provide semantic information for large-scale target detection. In this article, both low- and high-level features are taken into consideration to improve the accuracy of HSI change detection. In order to investigate the impact of depth of the network, we train five models by using different levels from 1 to 5 on the Jiangsu dataset. As shown in Fig. 10,

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TABLE III
COMPUTATIONAL COST OF DIFFERENT METHODS ON THE JIANGSU DATASET

Method	CVA	PCA	IR-MAD	SVM	ReCNN	BCNNs	DMPs^{2}raN
#Params	-	-	-	-	3850.9K	746.1K	631.7K
Running Time (s)	0.47	1.61	1.36	78.38	67.83	141.21	332.65

the model with two levels achieves the best performance in terms of OA and KC. As the number of levels increases, the performance of the model levels off or even degrades. It can be seen from the line chart that the models with three, four, and five levels are inferior to the proposed method. This is because the proposed model with two levels can already finely decompose the input image, and the learning problem has been well simplified. Furthermore, the network suffers from an overfitting problem when the number of model parameters is much too large.

3) Computational Efficiency: The running times and the number of parameters of different methods on the Jiangsu dataset are shown in Table III, and all the methods are tested on a desktop with NVIDIA GeForce GTX 2080Ti GPU. It can be observed that the running times of traditional methods are satisfactory at the cost of poor performance with many white noises. The CNN-based methods achieve better performance but are more time-consuming. The performance of the proposed DMPs^{2}raN is significantly improved with an acceptable time increment. Although the calculation burden is relatively heavy, the 3-D convolution helps the DMPs2raN fully extract the spatial–spectral feature of HSI, improving the accuracy of change detection. In order to reduce the redundant parameters of the model, we adopt 3-D depthwise separable convolution to remove the standard 3-D convolution. Given the input feature map of size $h \times w \times b \times c_i$, the standard 3-D convolutional layer generates a $h \times w \times b \times c_o$ feature as the output, where h, w, and b are the height, weight, and depth of feature map, and c_i and c_o represent the number of input channels and output channels, respectively. Note that the height, width, and band of HSI are corresponding to the height, width, and depth of 3D convolution. The number of trainable parameters of a standard 3-D convolutional layer with kernel size k is $k \times k \times k \times c_i \times c_o$. The 3-D depthwise separable convolution factorizes the standard 3-D convolution into a $k \times k \times k$ depthwise convolution and a $1 \times 1 \times 1$ pointwise convolution, which reduces the redundant parameters of the model. The number of trainable parameters of the 3-D depthwise separable convolution with the kernel size k is $k \times k \times k \times c_i + c_i \times c_o$. Furthermore, the bottleneck structure is adopted to compact the significant information of each scale, which improves the computational efficiency. As shown in Table III, the proposed method achieves the best performance with fewer parameters than both ReCNN and BCNNs. In our future work, we will investigate some more innovative methods to reduce the computational cost while ensuring performance.

4) Effectiveness of Multiscale Pyramid Convolution: Since the ground objects with multiplicative sizes can be captured by HSIs, the multiscale pyramid convolution is adopted to extract multiscale spatial–spectral features, which improves the feature representation capability of the proposed DMPs^{2}raN. To confirm the validity of the multiscale pyramid convolution, comparison experiments are conducted with convolutions of different numbers of scales on the Jiangsu dataset. Fig. 11 shows the visual results of the models with convolutions from one to three scales, in which red represents false positive and green represents false negative. As shown in Fig. 11(a) and (b), missed detection and false detection phenomenon still exist in the enlarged region. The proposed method with multiscale pyramid convolution (kernel sizes of 3, 5, and 7) achieves the best results with more details, especially in the boundaries and complex changed regions. The results show that the proposed multiscale pyramid convolution is beneficial to make full use of the multiscale features of HSIs, which is critical to improve the accuracy of HSIs' change detection.

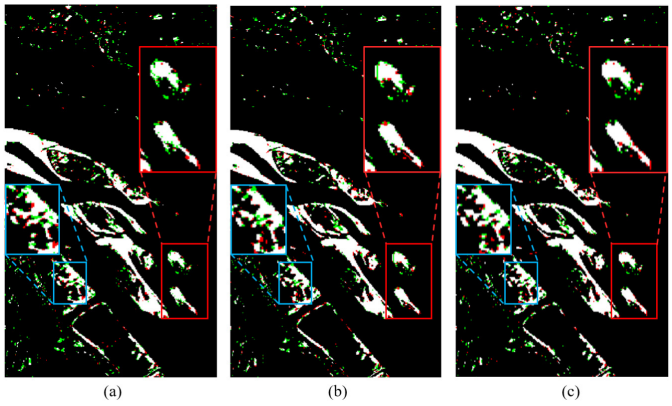


Fig. 11. Visual results of models with different scales of convolutions on the Jiangsu dataset. (a) Kernel size = 3. (b) Kernel size = 3, 5. (c) Kernel size = 3, 5, 7.

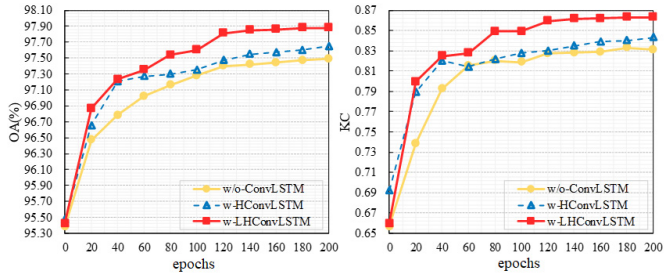


Fig. 12. Detection performance curves of models with and without ConvLSTM on the Jiangsu dataset.

5) Effectiveness of ConvLSTM: In the proposed DMPs^{2}raN, the ConvLSTM is adopted to model the temporal features between the HSIs without the loss of spatial correlation

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information. To investigate the influence of ConvLSTM, we perform the experiments on models, which removes the ConvLSTM (w/o-ConvLSTM), only feeds high-level features into ConvLSTM (w-HConvLSTM), and feeds both low- and high-level features into ConvLSTM (w-LHConvLSTM). The results of different models on the Jiangsu dataset are shown in Fig. 12. The OA and KC of model w/o-ConvLSTM are significantly lower than those of other models. The model w-HConvLSTM is superior to the model w/o-ConvLSTM with OA of 97.65% and KC of 0.8432, which indicates that the ConvLSTM contributes to improve the HSIs' change detection performance. Compared with the model w-HConvLSTM, the model w-LHConvLSTM increases the accuracy by 0.23% of OA and 0.0198 of KC. The results show that the proposed method, which feeds both low- and high-level features into ConvLSTM, fully extracts the temporal features of the input HSIs.

IV. Conclusion

In this article, a novel DL-based network called DMPs^{2}raN is proposed for HSI change detection. DMPs^{2}raN extracts multiscale features in both low and high levels to make full use of abundant information in HSIs. In order to further improve the feature representation capability, the S^{2}RAM is designed to highlight the informative features at each scale. The spatial–spectral features are fed into the ConvLSTM network in parallel for temporal feature extraction. Moreover, the proposed network can be trained in an end-to-end manner. The experimental results demonstrate that the proposed method is superior to the start-of-the-art methods with high effectiveness and reliability for HSI change detection.

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